Antonin LECOCQ

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Third-year Mathematics Student at Université Paris Dauphine – PSL

Education

- **Université Paris Dauphine – PSL** Sep. 2024 – Present
  Bachelor's Degree in Applied Mathematics *Paris, France*
  - **Coursework** Calculus, Algebra, Probability Theory, Algorithms and Programming, English...
- **EIB Étoile High School** Sep. 2021 – Jun. 2024
  French General Baccalaureate with high honors *Paris, France*
  - **Specializations** Mathematics, Physics and Chemistry, and Computer Science.

Experience

- **Astrophysics Research Intern – UROP PSL** Nov. 2026 – Present
  Observatoire de Paris – PSL *Paris, France*
  - Analysis of the Praesepe open cluster using Gaia DR4 data, membership determination.
- **Mathematics Research Intern** Jun. 2026 – Jul. 2026
  MAP5, Université Paris Cité *Paris, France*
  - Study of $SO(n)$ Lie group, reviewed English-language literature, and wrote a research report. Developed geometric registration models using Python and PyTorch.
- **Backend Developer Intern** Jul. 2025
  Celesta *Lisbon, Portugal*
  - Developed Symfony APIs within a microservices architecture, refactored PHP code, and worked with Docker. Designed and implemented new KYC functionality.
- **Treasurer & Webmaster** Oct. 2024 – May 2026
  Bureau des Arts de Dauphine *Paris, France*
  - **Treasurer** : Financial management of artistic projects, fundraising and event organization
  - **Webmaster** : Security, maintenance, and updates of three websites
- **Private Math Lesson *(Oct. 2026 – Present)* – Paris, France**
- **Sea Rescuer *(Aug. 2022)* – Cavalaire-sur-Mer Tourist Office, France**
- **Waiter *(Jul. 2021)* – Taokan, Paris, France**

Research

Lecocq, Antonin (2026). *Study of the Lie group SO(n) in the context of shape analysis*. Tech. rep. Under the supervision of Rayane Mouhli. Université Paris Cité, MAP5. DOI: 10.5281/zenodo.21557282.

Technical Skills

Programming	Python, C, PHP, Haskell, Bash Unix, R, LaTeX
Tools & Libraries	Git, Unix/Linux, Microsoft Excel, NumPy, Matplotlib, PyTorch
Languages	French, English, Italian

Certifications & Interests

GQS Trainer & PSC First Aid, French Red Cross May 2025
Level 2 Diver, FFESSM Aug. 2023
Recreational Boat License, Coastal Option, French Ministry of the Sea May 2023
Aeronautical Initiation Certificate (BIA), Ministry of National Education Jun. 2022
Advanced Open Water Diver, PADI Mar. 2022

First Place, at the Globeducate France Robotics Challenge – 2023
Ranked #6868 among 100,000+ users on Root-Me, with 1,980 points
Interests: Digital music composition, hiking, and scuba diving.